

Titanic Dataset Analysis

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Dataset Overview

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked	
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S
...	...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN	S
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42	S
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN	S
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148	C
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN	Q

Data Understanding & Preprocessing

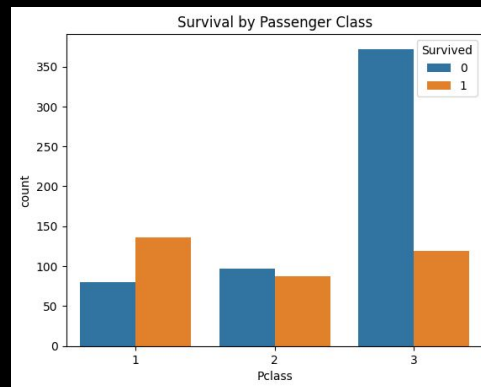
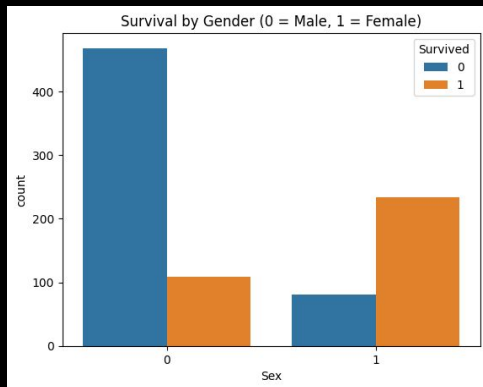
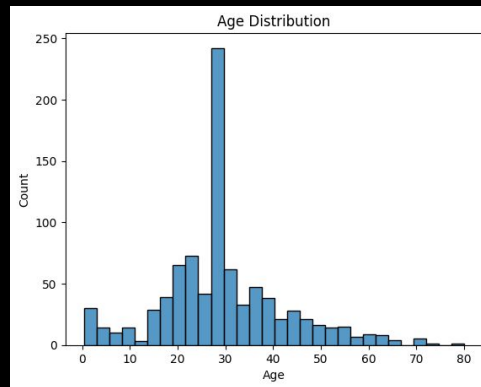
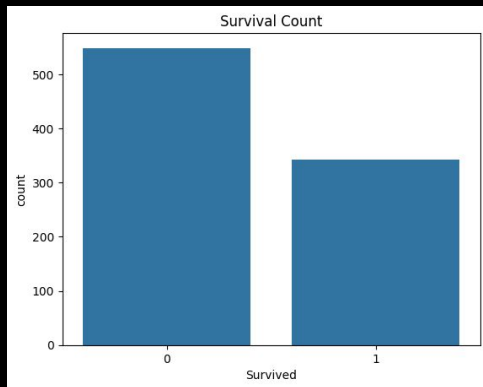
Missing values were handled and treated properly. Dataset was easier to interpret.

	Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked_Q	Embarked_S
0	0	3	0	22.0	1	0	7.2500	False	True
1	1	1	1	38.0	1	0	71.2833	False	False
2	1	3	1	26.0	0	0	7.9250	False	True
3	1	1	1	35.0	1	0	53.1000	False	True
4	0	3	0	35.0	0	0	8.0500	False	True

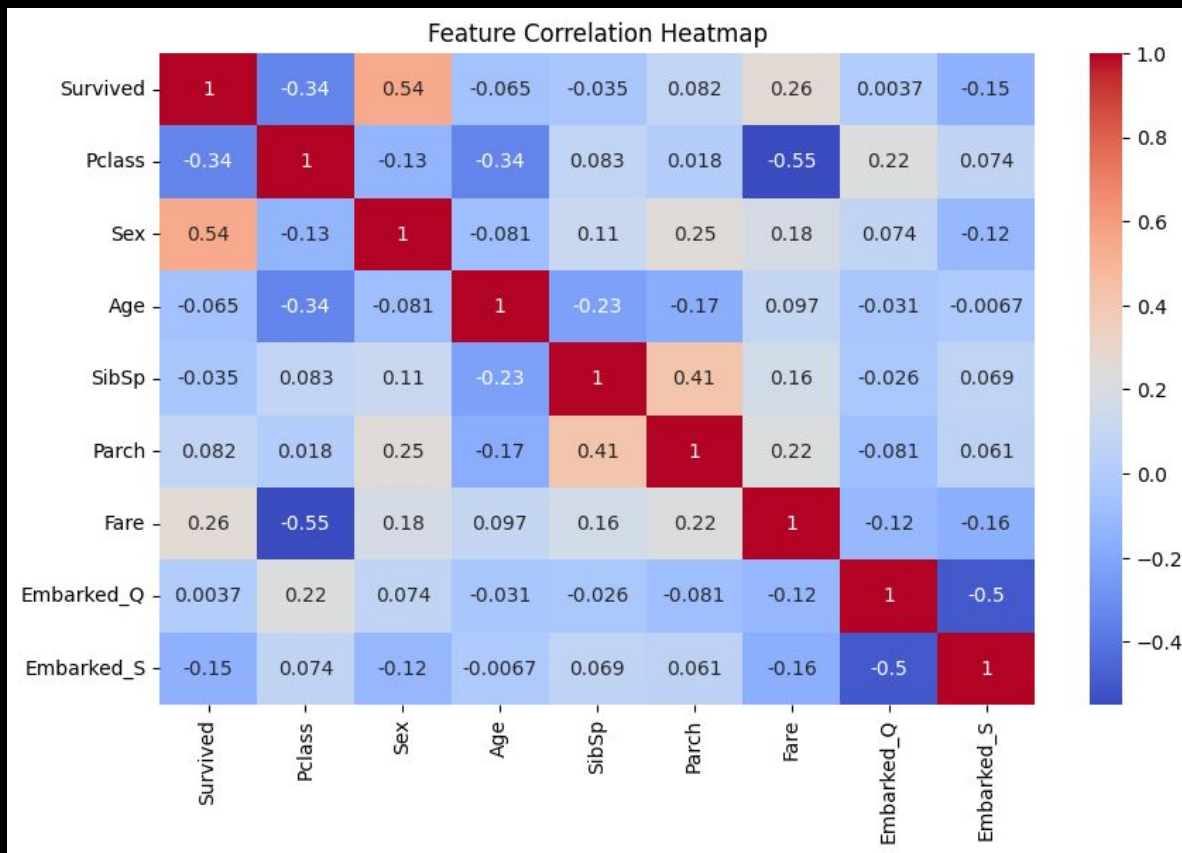
Missing Values:

```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age             177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin           687
Embarked         2
dtype: int64
```

Exploratory Data Analysis



Exploratory Data Analysis (cont.)



Model Setups

```
# Predictions & Evaluation
y_pred_log = log_model.predict(X_test)

print("=== Logistic Regression ===")
print("Accuracy:", accuracy_score(y_test, y_pred_log))
print("\nClassification Report:\n", classification_report(y_test, y_pred_log))
```

```
=== Logistic Regression ===
Accuracy: 0.8100558659217877
```

Classification	Report:				
	precision	recall	f1-score	support	
0	0.83	0.86	0.84	105	
1	0.79	0.74	0.76	74	
accuracy			0.81	179	
macro avg	0.81	0.80	0.80	179	
weighted avg	0.81	0.81	0.81	179	

```
# Predictions & Evaluation
y_pred_tree = tree_model.predict(X_test)

print("=== Decision Tree ===")
print("Accuracy:", accuracy_score(y_test, y_pred_tree))
print("\nClassification Report:\n", classification_report(y_test, y_pred_tree))
```

```
=== Decision Tree ===
Accuracy: 0.7877094972067039
```

Classification	Report:				
	precision	recall	f1-score	support	
0	0.82	0.82	0.82	105	
1	0.74	0.74	0.74	74	
accuracy			0.79	179	
macro avg	0.78	0.78	0.78	179	
weighted avg	0.79	0.79	0.79	179	

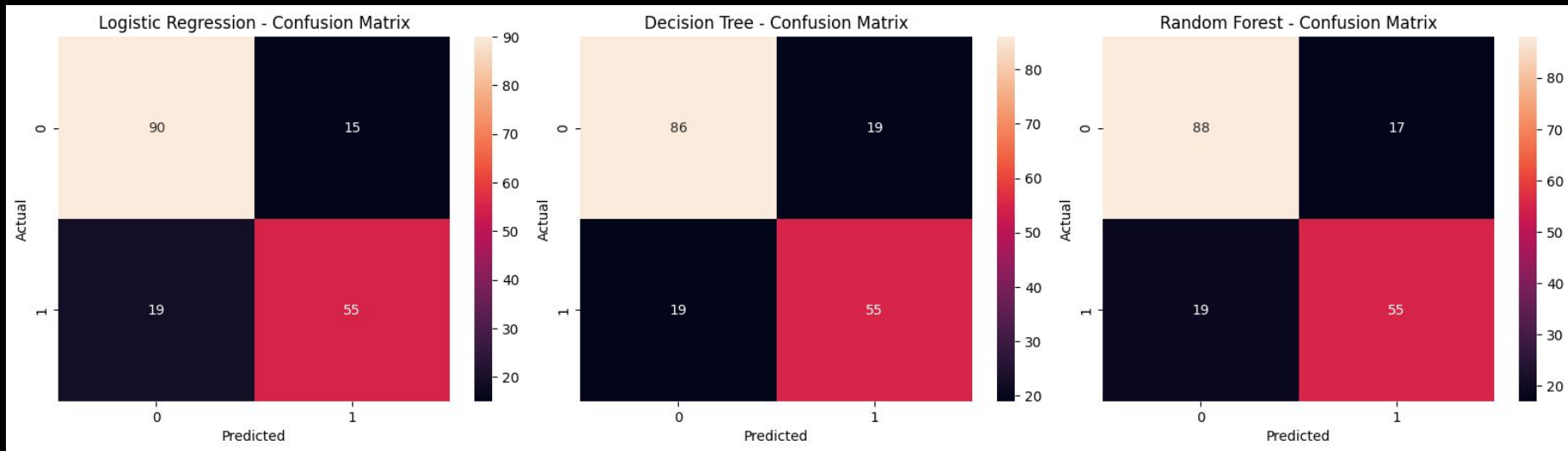
```
# Predictions & Evaluation
y_pred_rf = rf_model.predict(X_test)

print("=== Random Forest ===")
print("Accuracy:", accuracy_score(y_test, y_pred_rf))
print("\nClassification Report:\n", classification_report(y_test, y_pred_rf))
```

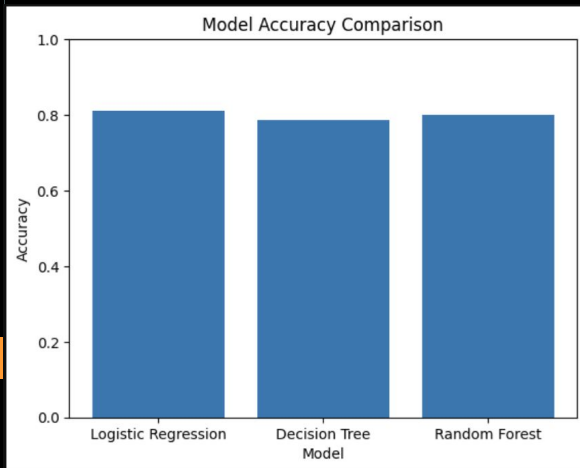
```
=== Random Forest ===
Accuracy: 0.7988826815642458
```

Classification	Report:				
	precision	recall	f1-score	support	
0	0.82	0.84	0.83	105	
1	0.76	0.74	0.75	74	
accuracy			0.80	179	
macro avg	0.79	0.79	0.79	179	
weighted avg	0.80	0.80	0.80	179	

Results/Model Comparison: Confusion Matrices



Model Accuracy Comparison



```
# Classification Report
print("Model Comparison:")
print("Logistic Regression Accuracy:", accuracy_score(y_test, y_pred_log))
print("Decision Tree Accuracy:", accuracy_score(y_test, y_pred_tree))
print("Random Forest Accuracy:", accuracy_score(y_test, y_pred_rf))
```

Model Comparison:
Logistic Regression Accuracy: 0.8100558659217877
Decision Tree Accuracy: 0.7877094972067039
Random Forest Accuracy: 0.7988826815642458

Feature Importance

```
# Logistic Regression (absolute coefficients)
log_importance = np.abs(log_model.coef_[0])

# Decision Tree importance
tree_importance = tree_model.feature_importances_

# Random Forest importance
rf_importance = rf_model.feature_importances_

features = X.columns

scaler = MinMaxScaler()

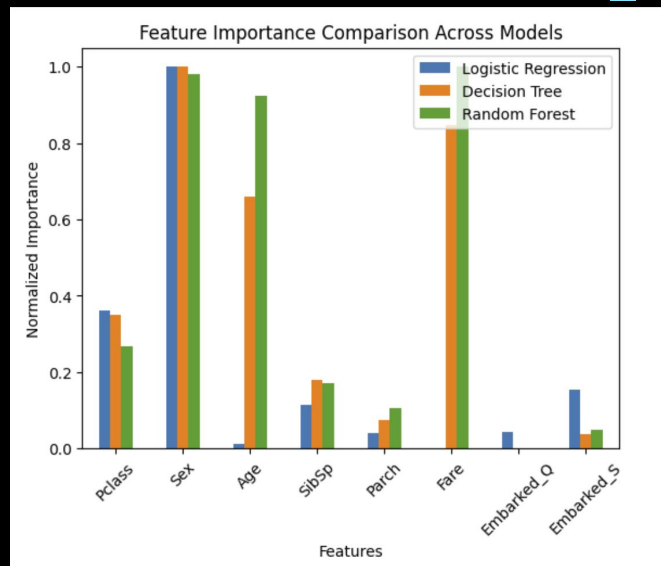
log_scaled = scaler.fit_transform(log_importance.reshape(-1,1)).flatten()
tree_scaled = scaler.fit_transform(tree_importance.reshape(-1,1)).flatten()
rf_scaled = scaler.fit_transform(rf_importance.reshape(-1,1)).flatten()

feat_compare = pd.DataFrame({
    'Feature': features,
    'Logistic Regression': log_scaled,
    'Decision Tree': tree_scaled,
    'Random Forest': rf_scaled
})

feat_compare = feat_compare.set_index('Feature')

feat_compare.plot(kind='bar')

plt.title("Feature Importance Comparison Across Models")
plt.ylabel("Normalized Importance")
plt.xlabel("Features")
plt.xticks(rotation=45)
plt.legend()
plt.show()
```



	Logistic Regression	Decision Tree	Random Forest
Feature			
Pclass	0.361388	0.350995	0.266534
Sex	1.000000	1.000000	0.980297
Age	0.010813	0.659373	0.923781
SibSp	0.113006	0.180049	0.169979
Parch	0.040776	0.074693	0.106163

Conclusion

Main findings:

- Gender was one of the strongest survival factors.
- Class, age, and fare also played an important role.
- Logistic Regression was our best model with about 81% accuracy.
- Feature importance showed that the models used sex, class, age, and fare the most.
- Overall, this project showed how passenger information can help predict survival and helped us practice data cleaning, graphs, and machine learning.

□ Kaggle Titanic Dataset: [kaggle.com/datasets/yasserh/titanic-dataset](https://www.kaggle.com/yasserh/titanic-dataset)